A Novel Antimony−Carbon Nanotube−Tin Oxide Thin Film: Carbon Nanotubes as Growth Guider and Energy Buffer. Application for Indoor Air Pollutants Gas Sensor

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A novel antimony−carbon nanotube−tin oxide (Sb−CNT−SnO_2) thin film using carbon nanotubes (CNTs) as growth guider and energy buffer was fabricated via the sol−gel method and ultrasonic dispersion technique, which could be potentially applied as a gas sensor for detecting indoor air pollutants emitted from building and decoration materials. The Sb−CNT−SnO_2 thin film shows higher sensitivity and shorter response time than the pure SnO_2 thin film and the Sb−SnO_2 thin film according to the gas sensing measurement. The possible mechanism of the better sensing performance has been primarily discussed from the aspects of growth guiding and energy buffering effects caused by CNTs. Furthermore, the electric field on the surface of Sb−CNT−SnO_2 thin film is proposed as the reason for rapid response to polar gases, e.g., formaldehyde, ammonia, and toluene.

1. Introduction

Tin oxide (SnO_2) as an n-type semiconductor with special electrical properties and excellent stability has been widely used as a gas sensing material.[1−3] Recently, in order to meet the strict demand for gas sensors in industry and daily life, more efforts have been made in this field,[4−6] and most of them focus on the enhancement of sensing performance by doping kinds of ions into SnO_2, such as Ni^{2+} and Zn^{2+},[7−9] and the preparation of nanoscale SnO_2 materials.[10,11] However, the gas sensing performance is still unsatisfactory for practical application in industry and civil due to limited sensitivity and poor selectivity so far.

Carbon nanotubes (CNTs) are considered as potentially applied materials for gas sensing because of their high specific surface area for adsorption,[12] nanoscale structure with numerous sites for chemical reactions,[13] and special metal or semiconductor properties with electrons storage and transport.[14,15] It was reported that CNTs, especially single-walled CNTs, and their incorporated materials with semiconductors had been investigated as sensing materials with competitive performance.[16,17] However, as is well-known, the single-walled CNTs are harmful to the semiconductivity of sensing materials because of their relatively low resistance.[18] In contrast, the multiwalled CNTs with poorer conductivity and more defects on the wall[19,20] may have more potential for the application in gas sensor.

Since the relationship between environment and health is regarded unprecedentedly, the cognition about the great impact on health by pollutants has been fixed in grain.[21] Generally, daily life indoors results in that we are surrounded by the indoor volatile pollutants, e.g., formaldehyde, ammonia, benzene, and toluene emitted from building and decorative materials, which are deleterious for our health with the high rate of cancer, respiratory disease, and baby abnormalities.[22−23] Therefore, the detection of these indoor air pollutants is of particular concern because of their long-term effects on our health, and the gas sensor with well-sensing performance is of great wealth for exploitation.

In this paper, a novel incorporated thin film of SnO_2, transition metal antimony (Sb), and multiwall CNTs was prepared via the sol−gel method and ultrasonic dispersion technique for the fabrication of indoor air pollutants gas sensor. For the facility of comparison on gas sensing properties among as-fabricated sensors, unlike some of our works researching on dynamic method previously,[24,25] the conventional static method was employed for the sensing tests toward formaldehyde, ammonia, benzene, and toluene as indoor air pollutants. The effect of CNTs as growth guider and energy buffer is proposed as the mechanism for the enhancement of gas sensing performance in the Sb−CNT−SnO_2 thin film. In addition, it is indicated that the electric field on the surface of thin film plays an important role, especially for detecting the polar gas molecules.

2. Experimental Details

2.1. Preparation of Thin Films. All the reagents and solvents, which were purchased from Shanghai Chemical Co., were analytical grade and used as received without further purification. Thin films were deposited on the $5 \times 2 \times 0.5$ mm^3 alumina (Al_2O_3) substrate with gold electrodes and ruthenium oxide (RuO_2) layer as heater for controlling the temperature, as shown in Figure 1. The tin(IV) chloride pentahydrate ($SnCl_4 \cdot 5H_2O$) dissolved in ethanol was employed as the precursor solution for the sol. After being mixed with antimony(III) chloride ($SbCl_3$) with molar percentage of 10 atom % to SnO_2, the mixed solution was refluxed with vigorous stirring by a magnetic stirrer for 6 h. Then, a certain amount of CNTs with weight percentage of 1−10 wt % to SnO_2, which had been prepurified in 0.5 M hydrochloric acid for 2 h, was dispersed

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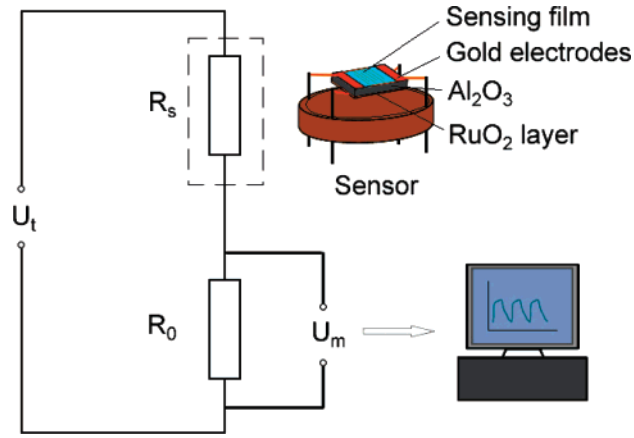

Figure 1. Schematic diagrams of the sensor and gas testing circuit.

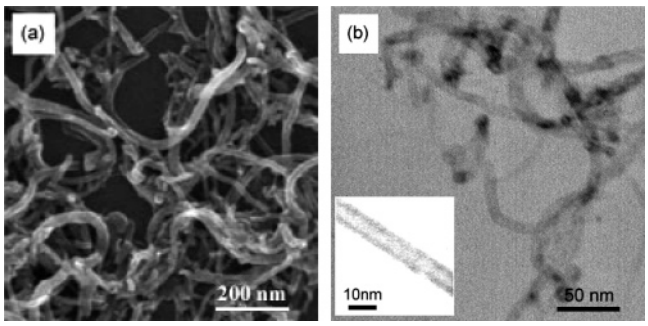

Figure 2. (a) FESEM micrograph and (b) TEM images of purified CNTs.

into the former solution completely by ultrasonic dispersing for 1 h. The CNTs were obtained via catalytic CVD approach with diameter from 6 to 8 nm and 3 to 5 layers, as shown in Figure 2. The final sol was dip-coated onto the substrate. The as-fabricated sensor samples were dried at 80 °C for 10 min, 150 °C for 20 min, and 80 °C for 10 min step by step for the initial formation of thin films without crack. Subsequently, the samples were calcined in air at 500 °C for 1 h. To control the thickness of thin film which was measured by a surface profilometer, the dip-coating and drying operations were carried out repeatedly.

2.2. Physical Characterization. The morphologies of CNTs and as-prepared thin films were investigated with a FEI Sirion 200 field emission scanning electronic microscopy (FESEM) and a Veeco Autoprobe CP atomic force microscopy (AFM), respectively. In order to obtain the information on adsorption and confirm the existence of CNTs in the tri-incorporated thin film, the Raman spectra were recorded with a Jobin−Yvon LABRAMHR spectrometer with argon ion laser beam focusing on the sample surface at room temperature. The crystalline structure analysis was performed with a Philips X'Pert Pro X-ray diffractometer (XRD) with Cu Kα radiation (1.5418 Å). The diffraction peaks of crystalline phases were compared with the standard patterns in JCPDS, and the crystalline size was estimated according to the Sherrer formula.[26] For the determination of the homogeneity and crystalline parameters, the transmission electron microscopy (TEM) and high-resolution TEM (HRTEM) were carried out with a JEOL 2010 transmission electron microscope. For the investigation on the energy effect, the UV−vis spectra were recorded with a Varian Cary-5E UV−vis−NIR spectrophotometer at wavelengths from 200 to 800 nm.

2.3. Gas Sensing Measurement. The gas sensing properties of thin films were investigated with a gas sensing measurement system, as shown in Figure 1. There is a gas container with dry air and target gases inputs which can be alternated for requirement. Meanwhile, a gas outlet is accessory. Besides, an adjustable potentiometer (R_0) is linked in the circuit as a voltage sharer with sensor (R_s), and the initial voltage on gas sensor can be adjusted for tests accordingly. Finally, the change of voltage during the gas test as the reflection of resistance change in sensor is acquired by a computer.

Formaldehyde, ammonia, benzene, and toluene as indoor air pollutants were tested, and their concentrations ranged from 100 to 1000 ppm. The sensitivity (S), as a function of gas sensing properties which was defined as the ratio of the resistance of sensor in air (R_{air}) to the one in target gas atmosphere (R_{gas}), could be calculated by the following expression:

$$S = \frac{R_{air}}{R_{gas}} = \frac{U_m''}{U_m'} \frac{U_t - U_m'}{U_t - U_m''} \quad (1)$$

where U_t was the total voltage of the testing circuit; U_m' and U_m'' were the voltages of R_0 in air or target gas, respectively. In addition, the response and recovery time were obtained by the time for 90% of final resistance change.

3. Results and Discussion

3.1. Surface Morphology and Microstructure of Thin Films. The surface morphologies of the sensing films with different incorporated materials are shown in Figure 3a−c. As presented in AFM images, the morphology of the SnO_2 thin film appears to be granular structure with narrow size distribution. The relatively smooth surface is transformed evidently into roughness and compaction by the incorporation with Sb and CNTs individually or simultaneously. The root-mean-square (rms) roughness of SnO_2, Sb−SnO_2, and Sb−CNT−SnO_2 thin films are 2.640, 3.015, and 3.902 nm, respectively. Furthermore, it can be observed that there are two typical profiles on the surface of Sb−CNT−SnO_2 thin films, namely, compact granular protuberances and array-like protrusions with height about 80 nm. The explanation for this phenomenon will be discussed later.

Figure 4 shows the XRD patterns of the SnO_2, Sb−SnO_2, and Sb−CNT−SnO_2 thin films. The diffraction peaks indicate that SnO_2 exists in the rutile structure (JCPDS No. 41-1445). Neither the Sb_2O_3 nor the Sb_2O_5 phase is detected in the patterns of Sb−SnO_2 and Sb−CNT−SnO_2. It is valuable for the appearance of $Sn_{0.918}Sb_{0.109}O_2$ diffraction peaks in Sb−SnO_2 and Sb−CNT−SnO_2 XRD patterns (JCPDS No. 88-2348), which indicates the existence of the composition as solid melts with two phases: one is SnO_2 and the other is $Sn_{0.918}Sb_{0.109}O_2$. In $Sn_{0.918}Sb_{0.109}O_2$ phase, the state of Sb is calculated to be Sb^{3+}. The calculated lattice parameters of SnO_2 by least-squares fit are $a = 0.4741$ nm and $c = 0.3189$ nm, which are in agreement with the reported values in JCPDS ($a = 0.4738$ nm and $c = 0.3187$ nm). However, the lattice parameters of Sb−SnO_2 and Sb−CNT−SnO_2, namely, $a = 0.4751$ nm, $c = 0.3195$ nm and $a = 0.4746$ nm, $c = 0.3193$ nm, respectively, are different from the values in JCPDS. It is ascribed to the lattice distortion caused by the substitution of Sn^{4+} (0.071 nm) by Sb^{3+} (0.078 nm). Moreover, the mean crystalline sizes of SnO_2, Sb−SnO_2, and Sb−CNT−SnO_2 thin films are 16.1, 7.9, and 9.3 nm, respectively, calculated from the Sherrer formula. It is indicated that the crystallites of SnO_2 are larger than the incorporated ones, which is consistent with some reports about the impact of dopants on crystallite size.[27,28] And this can be explained by Nae-Lih Wu's opinion;[29] i.e., because of the interaction on the boundaries between host and dopant crystallites, the motion of

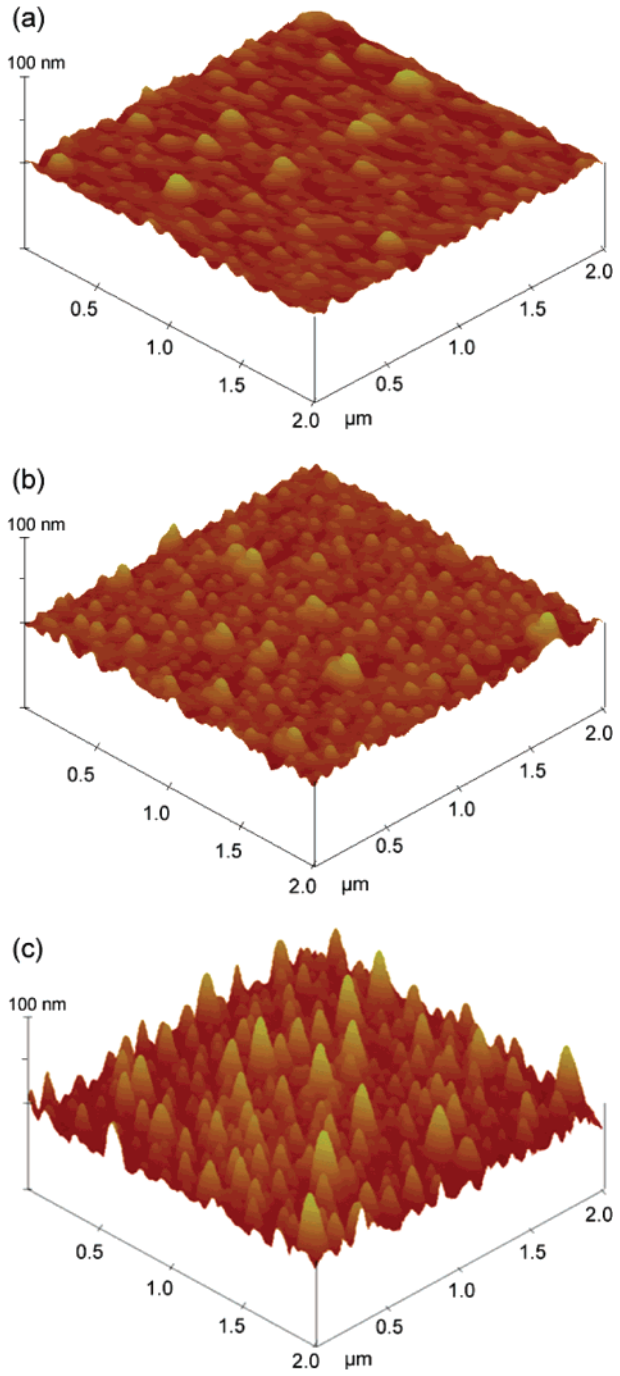

Figure 3. AFM images of (a) SnO_2, (b) Sb−SnO_2, and (c) Sb−CNT−SnO_2 thin films.

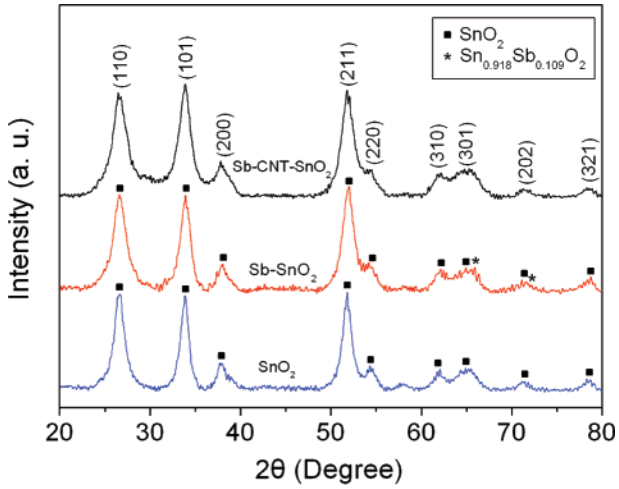

Figure 4. XRD patterns of the thin films with different incorporated materials.

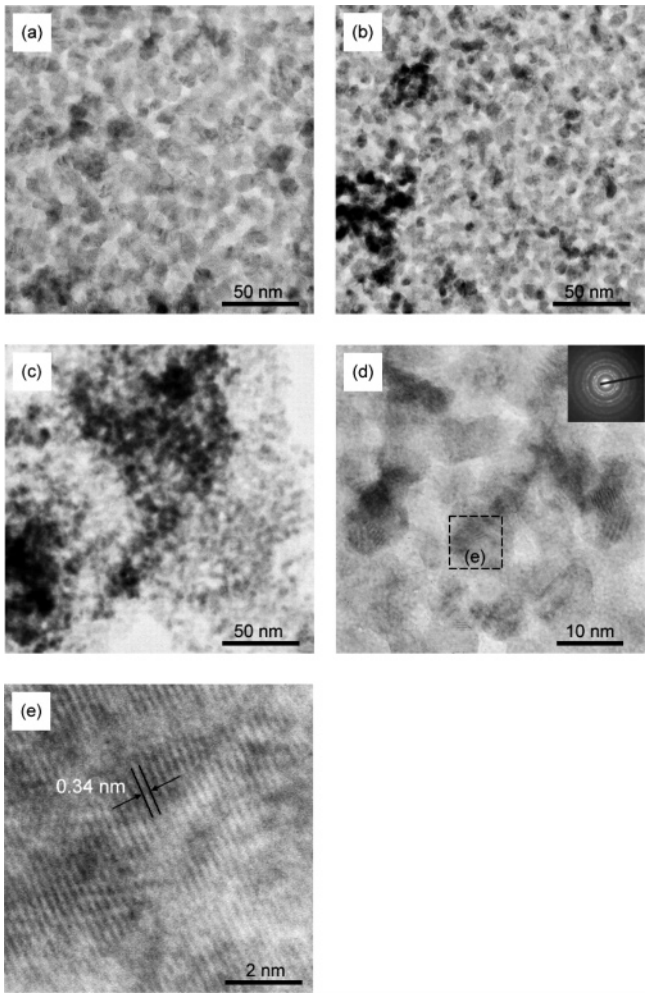

Figure 5. TEM micrographs of (a) SnO_2, (b) Sb−SnO_2, and (c) Sb−CNT−SnO_2 thin films and HRTEM images of (d) Sb−CNT−SnO_2 and (e) its magnification.

crystallites is restricted. In other words, the advancing of grain boundaries which is required for crystal growth is stunted. As a result, the size of crystallites is decreased by the doping of impurities.

In order to investigate the distribution and homogeneity of particles, the TEM and HRTEM micrographs of different thin films are shown in Figure 5a−e. The particle size of SnO_2 and Sb−SnO_2 is estimated to be 17 and 8 nm, respectively, which is consistent with the results calculated from XRD. On the contrary, the particles in the Sb−CNT−SnO_2 thin film with large distribution in size range from 7 to 12 nm are presented in Figure 5c,d. It is mainly caused by the different interaction in compact granular protuberances and array-like protrusions during the crystal growth process. The align lattice fringes of the Sb−CNT−SnO_2 are shown in Figure 5e, clearly reflecting the crystalline nature. The distance between two planes in crystal lattice is presented as 0.34 nm, which is consistent with the value of (110) planes according to the JCPDS.

The present experiment suggests that the CNTs cannot be observed in both TEM and HRTEM images of the Sb−CNT−SnO_2 thin film. Two reasons can be given for explaining this phenomenon. One is that the CNTs with light color are easily covered by the metal oxide particles with dark color, and the

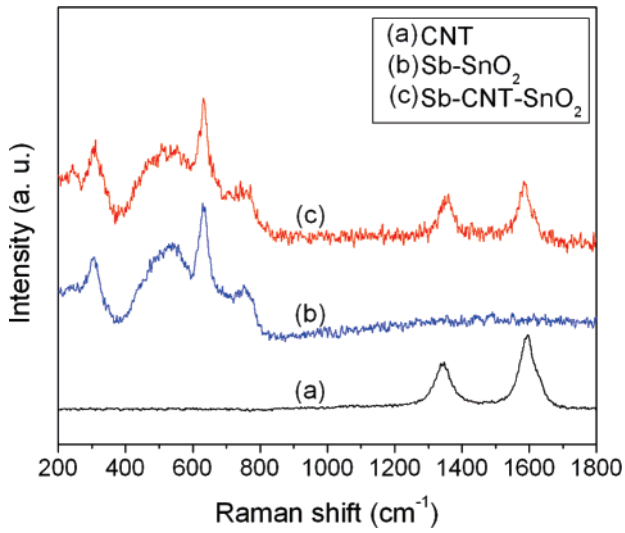

Figure 6. Room temperature Raman spectra of (a) CNTs, (b) Sb−SnO_2, and (c) Sb−CNT−SnO_2.

other one is that the CNTs have been lost in the preparing process of thin films. In order to support the explanation, the Raman spectra of CNTs, Sb−SnO_2 and Sb−CNT−SnO_2 have been obtained. As shown in Figure 6, the two peaks at 1360 and 1580 cm^{-1} in the Raman spectra of CNTs and Sb−CNT−SnO_2 are assigned as the Raman tangential modes in D-band and G-band of CNTs, respectively, which are the characteristics of CNTs.[30,31] On the contrary, they are absent in the Raman spectrum of Sb−SnO_2. Accordingly, the existence of CNTs in the Sb−CNT−SnO_2 thin film is confirmed. In addition, the peaks at 628 and 770 cm^{-1} are assigned to be the A_{1g} and B_{2g} vibration of tetragonal rutile SnO_2, which are the stretching vibrational modes of Sn−O bonds.[32] The peak at 315 cm^{-1} is the characteristic vibration of rutile structure of SnO_2 as well.[33] It is in good accordance with the XRD analysis. The peak at 570 cm^{-1} was reported as bonds in extrafine SnO_2 nanoparticles related to the small size effect.[34]

3.2. Gas Sensing Properties. Figure 7a−d shows the gas sensing response as thin films being exposed to target gases or dry air alternately. The concentration of the target gases, e.g., formaldehyde, ammonia, benzene, and toluene, is 500 ppm. It is presented that the voltage increases when the gas is injected and decreases as the input of dry air for the discharge of gas, as shown in Figure 7a. In other words, the resistance of sensing films decreases (from about 1 MΩ to 100 kΩ) as the atmosphere turned from dry air into target gas. In air surroundings, O_2 is adsorbed on the surface of thin films, and it will trap and react with the electrons from the sensing materials to produce negative oxygen ions, such as O_2^-, O^-, and O^{2-},[35] which set up an equilibrium state with each other demonstrated previously.[36] In contrast, when it is exposed to reductive gases, the reactions between negative oxygen ions and gas molecules will be happening, and the electrons are released in this process.[37] As a result, the high resistance in air is decreased in reductive gas surroundings.

It should be pointed out that the content of CNTs in the Sb−CNT−SnO_2 thin film is over 6 wt % because the influence on gas sensitivity of the Sb−CNT−SnO_2 thin films with different amounts of CNTs cannot be ignored. As shown in Figure 8, the sensitivity of the Sb−CNT−SnO_2 thin films with different amounts of CNTs is tested with gases at concentration of 500 ppm. As can be seen, the sensitivity is enhanced when the content of CNTs increases from 1 to 6 wt %. However, when

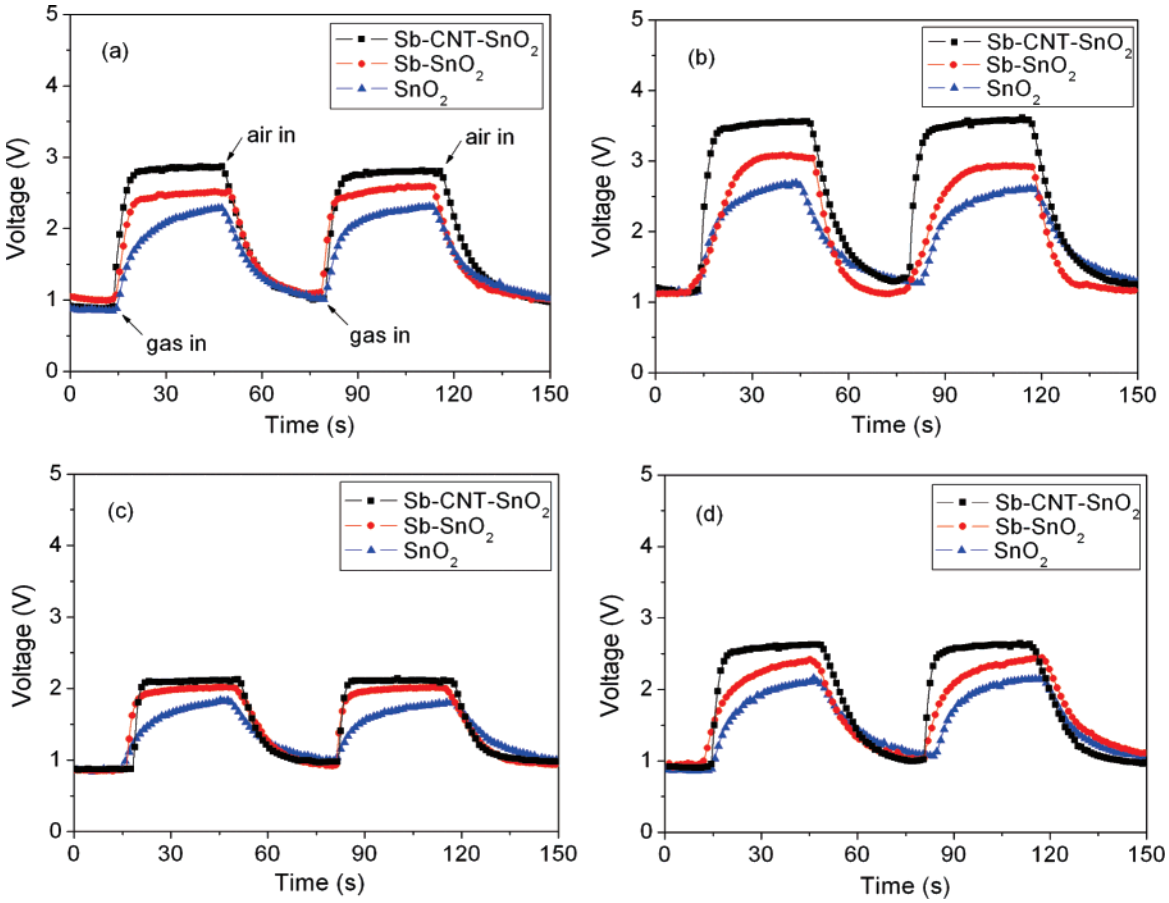

Figure 7. Gas sensing response of thin films to (a) formaldehyde, (b) ammonia, (c) benzene, and (d) toluene.

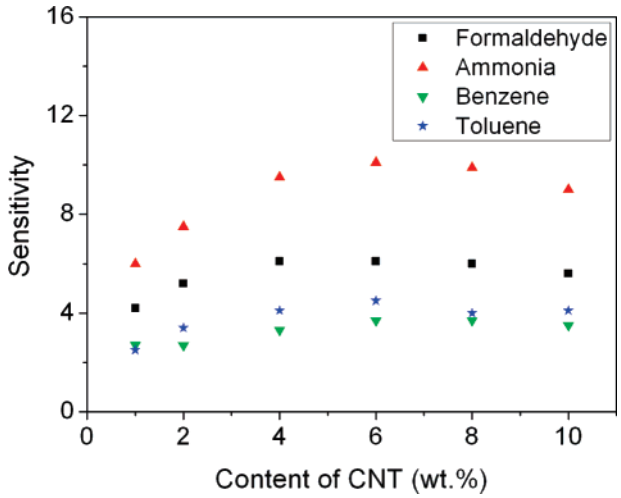

Figure 8. Gas sensitivity of Sb−CNT−SnO_2 thin films with different content of CNTs.

the content of CNTs is overfull, the sensitivity will be decreased due to the sharp decrease of resistance, which is caused by excess electric component inside the thin film.

The gas sensitivity of SnO_2, Sb−SnO_2, and Sb−CNT−SnO_2 thin films to formaldehyde, ammonia, benzene, and toluene at different concentrations is shown in Figure 9a−d. As presented, the sensitivity of the Sb−CNT−SnO_2 thin film is higher than the other two, especially to formaldehyde and ammonia. In the case of benzene, the sensing performances of Sb−SnO_2 and Sb−CNT−SnO_2 thin films are similar to each other. In contrast, the sensitivity of pure SnO_2 thin film is the lowest one. The response and recovery time as the two important functions of sensing properties are illustrated in Table 1. The response time of the Sb−CNT−SnO_2 thin film are significantly shorter than the other ones, especially to ammonia for 5 s, and the time of which is 13 and 16 s for SnO_2 and Sb−SnO_2, respectively. It can be also observed in Figure 7 with the acute peaks of the Sb−CNT−SnO_2 thin film.

3.3. Mechanism for Enhancement of Gas Sensing Properties by CNTs. As the differences of surface morphologies among thin films previously shown in AFM images, the crystallites are variously calculated on the surface, which is possibly caused by the crystal growth guiding effect of CNTs in the Sb−CNT−SnO_2 thin film. In compact granular protuberances, the crystallites collide mutually at the boundary tightly. At this condition, the growth of crystallites is limited, and the role as growth guider of CNTs is concealed. On the contrary, the impact on the crystal growth by CNTs in the array-like protrusions is dominant compared with the collision among crystallites themselves. Because of the strong adsorption of CNTs with large surface area, the initial nuclei for crystal growth are adsorbed on the surface of CNTs. As a result, the crystallites grow up predominantly along CNTs, and the protrusions are formed. The crystalline size is calculated from different crystalline planes from the XRD (Figure 4), and the deviation with the mean size is shown in Figure 10, where d_p and d_m are the sizes obtained from different crystalline planes and the mean one, respectively. From Figure 10, we calculated that the deviation of crystallite size in Sb−CNT−SnO_2 thin film is the greatest among the as-prepared three thin films. It implies that the crystal growth is oriented, which is consistent with the analysis on the surface morphology of the Sb−CNT−SnO_2 thin film.

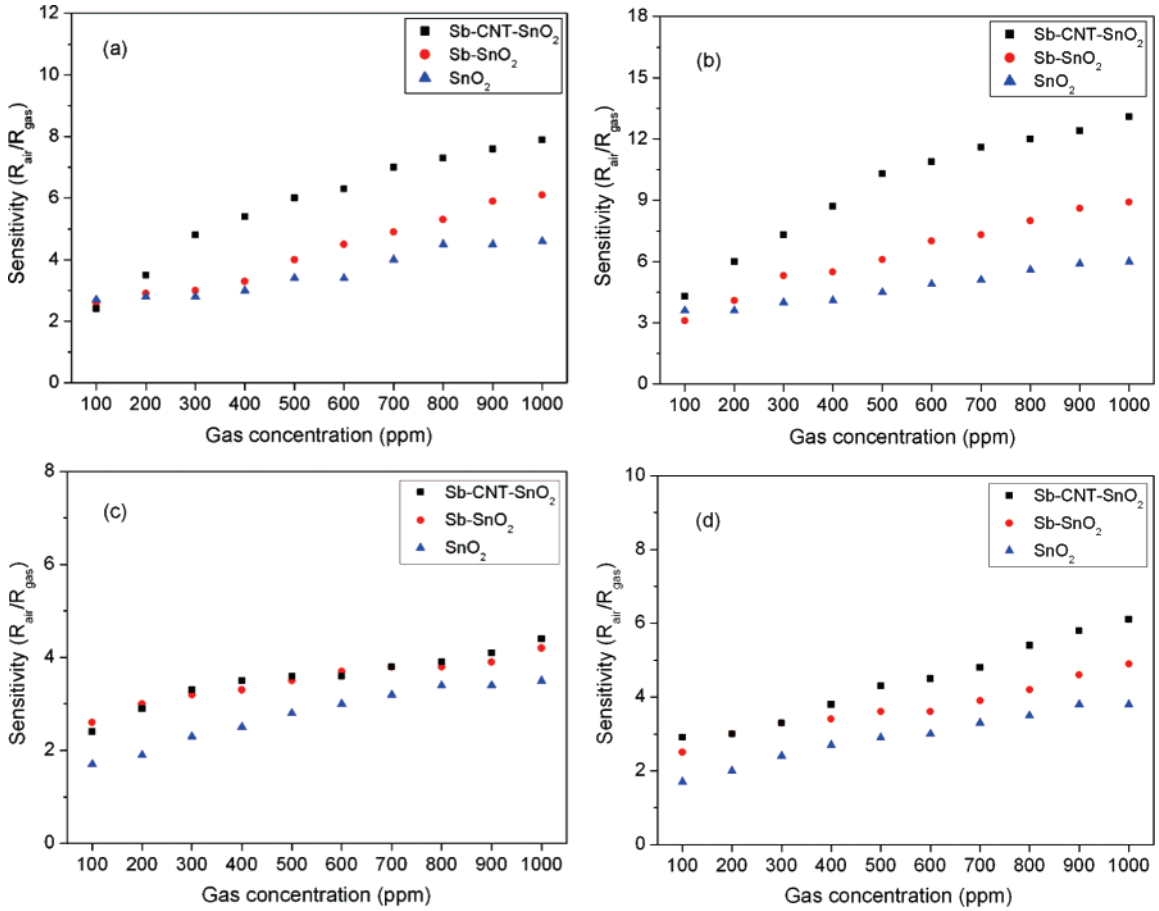

Figure 9. Gas sensitivity of thin films to (a) formaldehyde, (b) ammonia, (c) benzene, and (d) toluene at different concentration.

TABLE 1: Response and Recovery Time of Thin Films toward Gases at 500 ppm

	response time (s)			recovery time (s)		
	Sb−CNT−SnO_2	Sb−SnO_2	SnO_2	Sb−CNT−SnO_2	Sb−SnO_2	SnO_2
formaldehyde	7	7	16	18	18	23
ammonia	5	13	16	16	14	18
benzene	9	10	18	16	17	26
toluene	6	10	12	17	20	22

On the basis of the crystal growth guiding effect, it is proposed that an electric field exists on the surface of the Sb−CNT−SnO_2 thin film. In gas sensing tests, the enrichment of electric charge at the tip of array-like protrusions gives birth to the voltage grads, and then the electric field would be formed. As shown in Figure 11, this kind of electric field is consisted of numerous local electric fields. To our knowledge, voltage grads are the driving forces of the current. Accordingly, the sum of U_1, U_2, and U_n is U_0, which is used as the testing voltage. Because of polarity of gas molecule, such as formaldehyde, ammonia, and toluene, the adsorption can be accelerated by the electric field effectively. The enhanced transfer of gas molecules by the electric field force is favorable for both of the adsorption and reaction on the surface of sensing film. As a result, the response time of the Sb−CNT−SnO_2 thin film maybe shortened, especially to the polar gases as formaldehyde, ammonia, and toluene, except from nonpolar symmetrical benzene. As for the recovery time, the electric field cannot contribute to improving adsorption at the atmosphere of nonpolar O_2, and the performance on recovery time of the Sb−CNT−SnO_2 thin film is not quite different from the Sb−SnO_2 thin film.

Furthermore, as far as the energy effect inside the semiconductors is concerned, Figure 12a shows the transmission of SnO_2, Sb−SnO_2, and Sb−CNT−SnO_2 thin films. The absorption coefficient (α) can be calculated according to the formula 2, where T is transmission and d is the thickness of thin film.

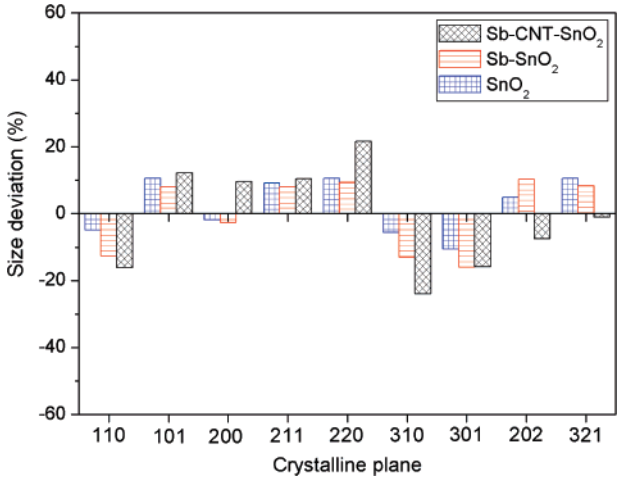

Figure 10. Deviation of crystalline size calculated from different planes toward mean size.

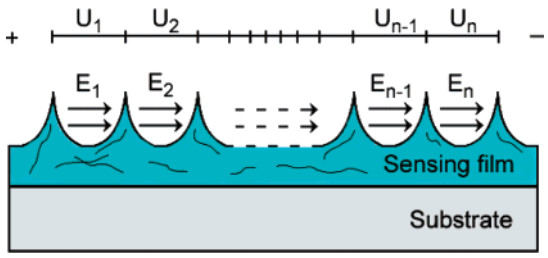

Figure 11. Schematic diagram of electric field on the surface of Sb−CNT−SnO_2 thin film.

Based on the relationship between α and the band gap energy (E_g),[38] in formula 3, where $h\nu$ is the photon energy (h and ν are the Planck constant and vibration frequency of photowave, respectively), the band gap of materials can be obtained. The plots of $(\alpha h\nu)^2$ vs $h\nu$ for different thin films are shown in Figure 12b.

$$\alpha = \frac{2.303 \log(1/T)}{d} \tag{2}$$

$$\alpha = A(h\nu - E_g)^{1/2} \tag{3}$$

As shown in Figure 12b, the band gap of pure SnO_2 film is 3.54 eV, which is agreement with the reported value (3.6 eV).[39] However, the band gap values of Sb−SnO_2 and Sb−CNT−SnO_2 films are 3.36 and 3.13 eV, respectively, which are lower than that of the pure one. A large numbers of oxygen vacancies (V_o) are created by the doping of Sb in the crystalline lattice of SnO_2 in the Sb−SnO_2 thin film; meanwhile, the defect energy caused by the impurity is strengthened.[40,41] As mentioned above, the doping impact in the Sb−CNT−SnO_2 thin film is more efficient by the more effective incorporation between Sb and

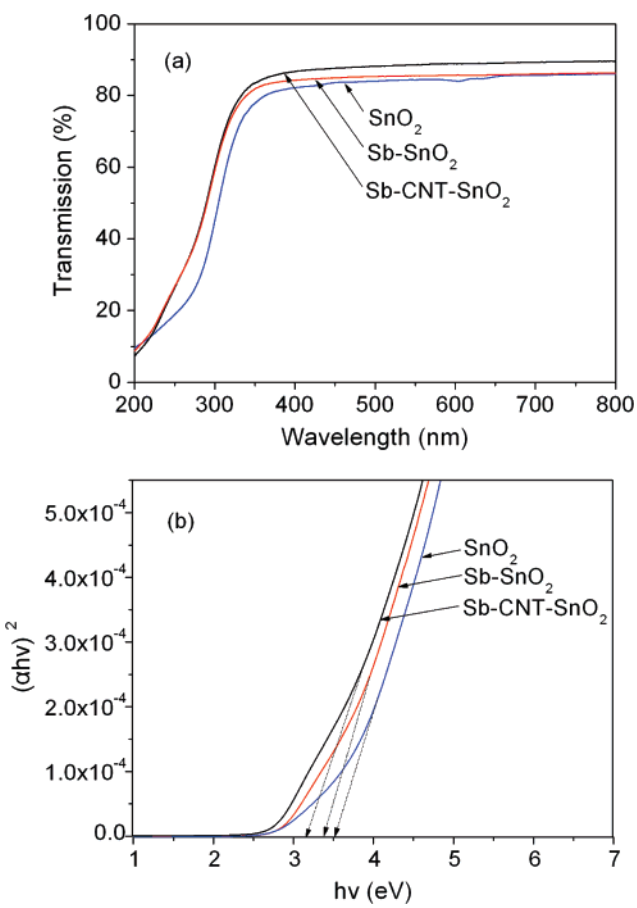

Figure 12. (a) UV−vis transmission spectra of SnO_2, Sb−SnO_2, and Sb−CNT−SnO_2 thin films. (b) Plots of $(\alpha h\nu)^2$ vs $h\nu$ for different thin films.

SnO_2 in the crystalline lattice, which resulted from their strong connections with CNTs. In addition, CNTs exhibit p-type semiconductor characteristics when being incorporated with metal oxides.[42] Hence, the existence of p−n junction in the Sb−CNT−SnO_2 thin film is reasonable due to the formation of p−n junction between p-type and n-type semiconductors in the incorporated materials.[43] Accordingly, it gives birth to the change of band gap. In other words, CNTs act as energy buffer in the Sb−CNT−SnO_2 thin film, which can decrease the high-energy requirement for the excitation of electrons from valence to conduct band. Consequently, more negative oxygen ions are produced in the reaction between adsorbed oxygen and electron. It is greatly favorable for the enhancement of gas sensing performance, which is a reflection of reactions between negative oxygen ions and gas molecules.

4. Conclusions

In conclusion, a novel Sb−CNT−SnO_2 thin film was prepared via the sol−gel method and ultrasonic dispersion technique for the application as gas sensor toward indoor air pollutants. Compared with SnO_2 and Sb−SnO_2 thin films, the morphology of Sb−CNT−SnO_2 thin film is obviously different. There are two typical profiles on the surface of Sb−CNT−SnO_2 thin films, e.g., compact granular protuberances and array-like protrusions. The mean size of crystallites in the tri-incorporated thin film is smaller than that of pure SnO_2 due to the inhibition of crystal growth by Sb; however, it is larger than the one of Sb−SnO_2 ascribed to the great vibration in size with different planes. On the basis of the gas sensing measurement, the Sb−CNT−SnO_2 thin film is found to give the best performance among them with high sensitivity and short response time, especially to formaldehyde, ammonia, and toluene. Because of the impact on semiconductivity by CNTs, the appropriate amount of CNTs in the Sb−CNT−SnO_2 thin film can achieve the optimal sensing properties. The crystal growth guiding and energy buffering effects are proposed as the mechanism of the enhancement of sensing properties by CNTs. Furthermore, on the surface of Sb−CNT−SnO_2 thin film with array-like protrusions, the existence of an electric field is proposed as the explanation for high-speed response to formaldehyde, ammonia, and toluene as polar molecules and ordinary performance to benzene with nonpolarity. It is believed that the Sb−CNT−SnO_2 thin film is wealthy for further research with great potential for application as gas sensor toward indoor air pollutants, and the electric field on the surface of sensing film is expected to be a new strategy for the improvement of gas selectivity.

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